

fix active: A LAMMPS Package for Ten Active-Matter Models

User Manual and Tutorial

Ram Chand

Saqia Bukhari

The Begum Nusrat Bhutto Women University, Sukkur, Pakistan

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1 Overview

`fix active` is a single LAMMPS fix that implements ten active-matter self-propulsion models inside a uniform input-script interface. Each model is a distinct `compute_<model>()` kernel registered in the fix's `post_force` callback; no kernel re-uses another model's update routine. The supported models are:

keyword	model
<code>abp</code>	Active Brownian Particles
<code>aoup</code>	Active Ornstein–Uhlenbeck Particles
<code>chiral</code>	Chiral ABP (intrinsic angular velocity ω_0)
<code>rtp</code>	Run-and-Tumble Particles
<code>vicsek</code>	Vicsek-type polar alignment
<code>rod</code>	Active rod with anisotropic friction
<code>spinner</code>	Chiral spinner with odd (Hall) viscosity
<code>iabp</code>	Inertial ABP
<code>mips</code>	Motility-Induced Phase Separation (Cates–Tailleur quorum sensing)
<code>nematic</code>	Active nematic with apolar Q -tensor torque

The current release applies the active force in the xy -plane (2D systems, or 3D boxes confined to a plane). Per-atom orientation, the AOUP active velocity, and a quaternion scaffold for a future 3D update are all registered for MPI atom exchange, ghost forward-communication, and the periodic atom-sort callback, so the framework is correct under arbitrary spatial domain decomposition.

2 Installation

2.1 Prerequisites

A working LAMMPS build environment. Any LAMMPS version from ~ 2022 onward is supported. No additional LAMMPS package is required.

2.2 Source files

Copy two files into the LAMMPS source tree:

```
cp src/fix_active.cpp $LAMMPS_ROOT/src/
cp src/fix_active.h   $LAMMPS_ROOT/src/
```

2.3 CMake build

```
cd $LAMMPS_ROOT
mkdir -p build && cd build
cmake -D BUILD_MPI=ON ../cmake
make -j8 lmp
```

2.4 Traditional Makefile build

From the LAMMPS `src/` directory:

```
make mpi -j8
```

2.5 Verification

Run any of the input scripts in `examples/` (or the benchmarks shipped with the manuscript repository). A successful invocation prints the banner line

`fix ID group active <model> ...` in the `log.lammps` echo of the input.

3 Common syntax

```
fix ID group active <model> keyword value [keyword value ...]
```

- **ID**: user-assigned fix identifier (LAMMPS standard).
- **group**: atom group to which the active force is applied.
- **model**: one of the ten keywords listed in Sec. 1.
- **keyword value pairs**: order-independent. Unrecognised keywords are flagged as errors; keywords that are recognised but not consumed by the chosen model are silently ignored (handy for sweeping a single template script over multiple models).

Global keywords that any model may accept (defaults in parentheses):

keyword	meaning	default
<code>v0</code>	self-propulsion speed	1.0
<code>Dr</code>	rotational diffusion coefficient	0.1
<code>Dt</code>	translational diffusion coefficient	0.0
<code>gamma</code>	friction γ used by the force prefactor	1.0
<code>kT</code>	thermal energy (for self-Langevin kernels)	1.0
<code>seed</code>	RNG seed (per-rank seed = <code>seed+rank</code>)	12345
<code>dimension</code>	override <code>domain->dimension</code> (rarely needed)	box dim.

Combination rules. The active force is added in `post_force`. Two of the kernels supply their own Stokes drag and fluctuation–dissipation noise and *must not* be combined with `fix langevin` or `fix brownian`:

- `rod` – self-contained anisotropic Langevin. Use with `fix nve` only.
- `iabp` – self-contained inertial Langevin. Use with `fix nve` only.

All other kernels (`abp`, `aoup`, `chiral`, `rtp`, `vicsek`, `spinner`, `mips`, `nematic`) add only the active force; for a complete overdamped Langevin integrator combine with `fix nve + fix langevin` (or `fix nve + fix brownian` on LAMMPS versions where it is available).

4 Models in detail

4.1 Active Brownian Particles (abp)

Overdamped self-propelled disks. Per-atom EoMs:

$$\begin{aligned}\dot{\mathbf{r}}_i &= v_0 \hat{\mathbf{e}}_i + \mu \mathbf{F}_i + \sqrt{2D_t} \boldsymbol{\xi}_i^t, \\ \dot{\theta}_i &= \sqrt{2D_r} \xi_i^r.\end{aligned}$$

The fix adds the force $\mathbf{F}_i^{\text{act}} = \gamma v_0 (\cos \theta_i, \sin \theta_i, 0)$, so the terminal speed is v_0 when combined with fix langevin with damping $\gamma_{\text{lang}} = \gamma$.

Keywords: v0 Dr [Dt] [gamma] [seed]

Long-time MSD (2D): $\langle \Delta r^2(t) \rangle \xrightarrow{t \gg \tau_r} 4D_{\text{eff}} t$, $D_{\text{eff}} = D_t + v_0^2/(2D_r)$, $\tau_r = 1/D_r$.

```
fix 1 active_atoms active abp v0 1.0 Dr 0.1 gamma 1.0 seed 54321
```

4.2 Active Ornstein–Uhlenbeck Particles (aoup)

Replaces the discrete orientation by a coloured-noise active velocity $\mathbf{v}_i^a$:

$$\dot{\mathbf{r}}_i = \mu \mathbf{F}_i + \mathbf{v}_i^a, \quad \dot{\mathbf{v}}_i^a = -\tau^{-1} \mathbf{v}_i^a + \sqrt{2D_a/\tau} \boldsymbol{\eta}_i.$$

The kernel uses the *exact* OU propagator $\mathbf{v}_i^a(t + \Delta t) = e^{-\Delta t/\tau} \mathbf{v}_i^a(t) + \sqrt{D_a(1 - e^{-2\Delta t/\tau})} \mathbf{Z}_i$.

Keywords: Da tau [gamma] [seed]

ABP-equivalent parameters (2D): $\tau = 1/D_r$, $D_a = v_0^2/2$ gives the same long-time diffusion as ABP.

```
fix 1 active_atoms active aoup Da 0.5 tau 10.0 gamma 1.0 seed 54321
```

4.3 Chiral ABP (chiral)

ABP with deterministic angular drift ω_0 : $\dot{\theta}_i = \omega_0 + \sqrt{2D_r} \xi_i$. Trajectories are circular of radius $v_0/|\omega_0|$ at short times.

Keywords: v0 Dr omega [gamma] [seed]

Long-time diffusion (2D): $D_{\text{eff}}^{\text{chiral}} = D_t + v_0^2 D_r / [2(D_r^2 + \omega_0^2)]$.

```
fix 1 active_atoms active chiral v0 1.0 Dr 0.1 omega 0.5 gamma 1.0 seed 54321
```

4.4 Run-and-Tumble Particles (rtp)

Ballistic “runs” at v_0 punctuated by instantaneous tumbles drawn from a Poisson process of rate `tumble_rate`. No continuous angular diffusion.

Keywords: v0 tumble_rate [Dt] [gamma] [seed]

Long-time diffusion (2D): $D_{\text{eff}}^{\text{RTP}} = D_t + v_0^2/(d\alpha)$, d = spatial dimension.

```
fix 1 active_atoms active rtp v0 1.0 tumble_rate 0.1 gamma 1.0 seed 54321
```

4.5 Vicsek-type alignment (vicsek)

Each particle aligns toward the circular mean of its neighbours within a cutoff `Rcut`. The blend between own axis and neighbour mean is done on the unit circle (well-defined across $\pm\pi$):

$$\begin{aligned}(\cos \theta_i, \sin \theta_i) &\leftarrow (1 - \alpha) (\cos \theta_i, \sin \theta_i) + \alpha (\overline{\cos \theta}, \overline{\sin \theta}), \\ \theta_i^{\text{new}} &= \text{atan2}(\sin_{\text{blend}}, \cos_{\text{blend}}) + \sqrt{2D_r \Delta t} \xi_i.\end{aligned}$$

With $\alpha = 1$ (default) this recovers the standard Vicsek mean.

Keywords: v0 Dr Rcut [alignment] [gamma] [seed]

Parallel note: per-atom θ is forward-communicated to ghosts, so the alignment kernel is correct under spatial decomposition.

```
fix 1 active_atoms active vicsek v0 1.0 Dr 0.1 Rcut 1.0 alignment 1.0 gamma 1.0 seed
54321
```

4.6 Active rod (rod)

Anisotropic-friction self-propelled rod with self-contained Langevin integrator. Parallel/perpendicular frictions $\gamma_{\parallel} = \gamma/\text{aspect}$, $\gamma_{\perp} = \gamma$; the kernel applies its own $-\gamma_{\parallel,\perp}v$ drag, $\sqrt{2\gamma k_B T/\Delta t}$ noise, and a deterministic propulsion $\gamma_{\parallel}v_0\hat{\mathbf{e}}_i$. Aligning torque is computed from the external force present in `atom->f` at entry to `post_force`.

Keywords: v0 Dr aspect [kT] [gamma] [seed]

Combine with: fix nve only.

```
fix 1 rod_atoms active rod v0 1.0 Dr 0.1 aspect 2.0 gamma 1.0 kT 0.01 seed 54321
```

4.7 Spinner with odd viscosity (spinner)

Adds the Hall force $\mathbf{F}_{\text{odd}} = \eta_{\text{odd}}\omega_{\text{spin}}(\hat{\mathbf{z}} \times \mathbf{v})$. The frictional drag γ that enters the Hall-angle prediction $\tan \phi = \eta_{\text{odd}}\omega_{\text{spin}}/\gamma$ comes from a companion `fix langevin` ($\gamma = m/\text{Tdamp}$); the spinner kernel does not duplicate that drag.

Keywords: omega_spin eta_odd [seed]

Combine with: fix nve + fix langevin.

```
fix nve_int spin_atoms nve
fix drag spin_atoms langevin 0.0 0.0 1.0 99999 # Tdamp = 1 -> gamma = 1
fix active1 spin_atoms active spinner omega_spin 1.0 eta_odd 1.0 seed 54321
```

4.8 Inertial ABP (iabp)

Massive ABP with self-contained inertial Langevin: $m\ddot{\mathbf{r}}_i = -\gamma\dot{\mathbf{r}}_i + \gamma v_0\hat{\mathbf{e}}_i + \sqrt{2\gamma k_B T}\boldsymbol{\xi}_i$. Stokes number $\text{St} = mD_r/\gamma$.

Keywords: v0 Dr gamma kT [seed]

Combine with: fix nve only.

```
fix 1 iabp_atoms active iabp v0 1.0 Dr 0.1 gamma 10.0 kT 0.1 seed 54321
```

4.9 Motility-Induced Phase Separation (mips)

Cates–Tailleur quorum-sensing realisation: the swim speed depends on the local density,

$$v(\rho_i) = v_0 \max(1 - \rho_i/\rho_{\text{max}}, 0).$$

The local density ρ_i is estimated from the neighbour count inside a ball of radius `Rcut`.

Keywords: v0 Dr Rcut rho_max [gamma] [seed]

Combine with: fix nve + fix langevin + pair_style soft/lj/cut.

```
pair_style lj/cut 1.122462
pair_coeff * * 1.0 1.0 1.122462
fix 1 mips_atoms active mips v0 30.0 Dr 0.05 Rcut 1.5 rho_max 1.0 gamma 1.0 seed 54321
```

4.10 Active nematic (nematic)

Dry, particle-based active nematic. Local Q -tensor over neighbours within R_{cut} , director angle $\phi_i = \frac{1}{2} \arctan(2Q_{xy}/Q_{xx})$, apolar alignment torque $-K \sin 2(\theta_i - \phi_i)$, and polar self-propulsion of amplitude `activity` along the body axis.

Keywords: activity K Rcut [Dr] [gamma] [seed]

Combine with: `fix nve + fix langevin + pair_style` for the WCA-style steric repulsion.

Parallel note: per-atom θ forward-communicated to ghosts.

```
fix 1 nematic_atoms active nematic activity 1.0 K 1.0 Rcut 1.5 Dr 0.05 gamma 1.0 seed
54321
```

5 Parallel computing

The per-atom angle θ_i , the AOUP active velocity $\mathbf{v}_i^a$, and the quaternion scaffold all follow the atom across MPI exchange (via overridden `pack_exchange` / `unpack_exchange`) and through the periodic atom-sort that LAMMPS performs when rebuilding neighbour lists (`copy_arrays`). In addition, the Vicsek and nematic kernels forward-communicate θ_i to ghost atoms at the start of each `post_force` invocation via `comm_forward = 1` and overridden `pack_forward_comm` / `unpack_forward_comm`. This is critical: without it, alignment partners across every subdomain boundary (*including* periodic-image neighbours in serial runs) are silently dropped, and the apparent Vicsek transition shifts to lower noise.

The kernels are not OpenMP-threaded. For hybrid MPI+OpenMP runs set `OMP_NUM_THREADS=1` for the active `fix` and parallelise across MPI ranks only.

6 Worked examples

6.1 Example 1 – ABP MSD validation

```
units          lj
dimension      2
boundary       p p p
atom_style     atomic
atom_modify    map array

variable       N          equal 1000
variable       L          equal 1000
variable       v0         equal 1.0
variable       Dr         equal 0.1
variable       kT         equal 0.01

region         box block 0 ${L} 0 ${L} -0.5 0.5
create_box     1 box
create_atoms   1 random ${N} 12345 box
mass           1 1.0
velocity       all set 0.0 0.0 0.0
pair_style     none

fix nve        all nve
fix lang       all langevin ${kT} ${kT} 1.0 11111
fix active1    all active abp v0 ${v0} Dr ${Dr} gamma 1.0 seed 54321
fix enf2d      all enforce2d

compute        msd all msd com no
```

```
thermo_style custom step temp c_msd[4]
thermo      5000

fix          msdout all print 1000 "$(step*dt) $(c_msd[4])" file msd.dat &
            screen no title "# t MSD"

timestep     0.001
run          1000000
```

Expected long-time slope of the MSD = $4D_{\text{eff}} = 4(D_t + v_0^2/(2D_r)) \approx 20.04$ with $D_t = k_B T/\gamma = 0.01$, $v_0 = 1$, $D_r = 0.1$.

6.2 Example 2 – Vicsek phase diagram under MPI

```
mpirun -np 4 lmp -in in.vicsek_phase_diagram
```

The decomposition is $2 \times 2 \times 1$ for a 2D box. Across each subdomain boundary the new forward-comm path supplies θ on ghost atoms; the transition is then in $D_r \in [25, 50]$ for $N = 500$, $L = 30$, $R_c = 1$.

6.3 Example 3 – Spinner Hall angle

```
fix nve_int  spin nve
fix drag     spin langevin 0.0 0.0 1.0 99999          # gamma_lang = 1
fix drive    spin addforce 1.0 0.0 0.0              # F_ext = (+1, 0)
fix active1  spin active spinner omega_spin 1.0 eta_odd 1.0 seed 54321
compute vx   spin reduce ave vx
compute vy   spin reduce ave vy
thermo_style custom step temp c_vx c_vy
```

Steady state: $\langle v_x \rangle = \langle v_y \rangle = \gamma F / [\gamma^2 + (\eta_{\text{odd}} \omega)^2] = 0.5$, $\tan \phi = v_y / v_x = 1$.

7 Diagnostics and dump output

`fix active` does not declare any extra per-atom or scalar outputs, so the standard `dump` `custom`, `compute msd`, `compute coord/atom`, `compute reduce`, etc. all work without modification on the simulated trajectory. If you need the per-atom orientation in a dump, add a `compute property/atom` of `ix iy iz` (positions) and dump the velocity to recover $\theta_i = \text{atan2}(v_y, v_x)$ in post-processing.

8 Performance

The active force is purely local for every kernel except Vicsek and nematic, which carry one extra forward-communication of one double per atom per timestep. In a typical benchmark (10000 ABP particles with WCA interactions, single CPU core), `fix active`'s `post_force` contribution is $\sim 9\%$ of the total wall time, dominated by the pair-force computation ($\sim 62\%$) and the neighbour-list rebuild ($\sim 18\%$).

9 Troubleshooting / FAQ

“Illegal fix active command: unknown keyword ...” A keyword in the input deck is not in the parser's whitelist. Check for typos; valid keywords are listed in the comment header of `src/fix_active.cpp`.

“fix active Rcut exceeds neighbor cutoff” The alignment / density cutoff is larger than the global LAMMPS neighbour cutoff. Increase the pair cutoff or the neighbour skin so that all neighbours within Rcut are included in the neighbour list.

“fix active currently propels in the xy-plane only” Warning emitted when the box is declared 3D. The active force has no z -component; particles are free to diffuse in z through whatever other fix you have attached.

Hall angle is twice what I expect for the spinner. You are double-counting the drag. `fix active spinner` does *not* add a Stokes term; the drag must come from `fix langevin`. If you also added `fix langevin` on top of an explicitly drag-supplying kernel, remove one of them.

MSD slope wrong by a factor of 2. Most likely the wrong dimensionality factor. The 2D long-time slope is $4D_{\text{eff}}$; the 1D slope is $2D_{\text{eff}}$. Also check that you are fitting in the diffusive regime $t \gg \tau_r$ – a fit window of $[5\tau_r, 50\tau_r]$ is recommended (well past the ballistic crossover but before finite- N noise dominates the late-time tail).

Different result on 1 vs 4 MPI ranks for Vicsek/nematic. You are running an older `fix_active.cpp` that does not forward-communicate θ to ghosts. Pull the current version from [the repository](#).

10 Citation

If you use `fix active` in published work please cite:

Ram Chand and Saqia Bukhari, *Active Matter Model Implementation in LAMMPS: A Coherent `fix active` Framework of Ten Active Self-Propulsion Mechanisms*, (manuscript, 2026).
Source: github.com/ramc77/fix-active-lammps.

11 License

`fix_active.cpp`/`fix_active.h` are distributed under the GNU General Public License v3, matching the host LAMMPS license.